

AI/ML Internship – Day 59 Notes & Tasks

Module 8: Building a Rule-Based Chatbot using Python

Part 1: Recap of Day 58

Yesterday we learned:

-  What is a Chatbot?
-  Chatbot Architecture
-  Rule-Based Chatbots
-  AI-Based Chatbots
-  NLP in Chatbots

Today we will build our **first chatbot** using Python.

Part 2: What is a Rule-Based Chatbot?

Definition

A Rule-Based Chatbot is a chatbot that responds using predefined rules.

It does **not learn** from users.

It simply checks the user's input and gives the matching response.

Simple Explanation

Think of it like a traffic signal.

If:

Red

↓

Stop

If:

Green

↓

Go

Similarly,

If user says:

Hello

↓

Bot says:

Hi! Welcome.

Everything is already programmed.

Part 3: How Does a Rule-Based Chatbot Work?

User Types Message

|



Program checks input

|



Matches predefined rule

|



Returns response

Part 4: Real-Life Example

Imagine a college enquiry chatbot.

Student:

What are the course fees?

Bot:

The course fee is ₹15,000.

Student:

Where is the college?

Bot:

Our college is located in Kochi.

Student:

Can I learn dancing?

Bot:

Sorry, I don't understand your question.

Part 5: Chatbot Logic

Every chatbot follows this logic.

User Message

↓

Compare with Rules

↓

Found?

↓

Yes → Send Response

↓

No

↓

Default Response

Part 6: Python Program (Using if-elif)

```
print("🤖 AI Assistant")  
print("Type 'bye' to exit.\n")
```

```
while True:

    user = input("You: ").lower()

    if user == "hello":
        print("Bot: Hello! How can I help you?")

    elif user == "how are you":
        print("Bot: I'm doing great!")

    elif user == "your name":
        print("Bot: I'm an AI Chatbot.")

    elif user == "bye":
        print("Bot: Goodbye!")
        break

    else:
        print("Bot: Sorry, I don't understand.")
```

Part 7: Program Output

You: hello

Bot: Hello! How can I help you?

You: your name

Bot: I'm an AI Chatbot.

You: bye

Bot: Goodbye!

Part 8: Understanding the Code

while True

Creates an infinite loop.

while True:

Meaning:

The chatbot keeps talking until the user exits.

input()

```
user = input("You: ")
```

Reads the user's message.

lower()

```
user.lower()
```

Converts everything into lowercase.

Example:

HELLO

↓

hello

So:

Hello

HELLO

hello

are treated the same.

if-elif-else

Checks different conditions.

Example:

```
if user == "hello":
```

If matched,

Send response.

Otherwise,

Check the next condition.

break

Stops the chatbot.

break

Used when the user types:

bye

Part 9: Using Dictionary Instead of if-elif

Instead of writing many if statements,

we can use a dictionary.

Example:

```
responses = {  
    "hello": "Hello!",  
    "hi": "Hi!",  
    "how are you": "I'm fine.",  
    "your name": "I'm AI Bot."  
}  
  
user = input("You: ").lower()  
  
print(  
    responses.get(  
        user,  
        "Sorry! I don't understand."  
    )  
)
```

Why Dictionary?

Advantages:

 Less code

✓ Easy to manage

✓ Faster lookup

📌 Part 10: Making Chatbot Continuous

```
responses = {  
    "hello":"Hello!",  
    "hi":"Hi!",  
    "how are you":"I'm fine.",  
    "your name":"I'm AI Bot."  
}
```

while True:

```
    user = input("You: ").lower()
```

```
    if user == "bye":  
        print("Bot: Goodbye!")  
        break
```

```
    print(  
        "Bot:",  
        responses.get(  
            user,  
            "Sorry! I don't understand."  
        )  
    )
```

📌 Part 11: Real-World Example

Banking Chatbot

User:

Check balance

↓

Bot:

Please login first.

Hospital Chatbot

User:

Book appointment

↓

Bot:

Please select a doctor.

College Chatbot

User:

Admission

↓

Bot:

Admissions are open now.

📌 Part 12: Advantages of Rule-Based Chatbots

- ✅ Easy to build
 - ✅ Fast
 - ✅ Beginner friendly
 - ✅ No training data needed
 - ✅ Works well for FAQs
-

📌 Part 13: Limitations

- ❌ Cannot understand new questions.

Example:

Rule:

hello

User writes:

hey

Bot may fail.

✗ No learning ability.

✗ Cannot understand context.

✗ Limited intelligence.

✦ Part 14: Difference Between Rule-Based and AI Chatbot

Rule-Based	AI Chatbot
Uses if-else or rules	Uses AI & NLP
No learning	Learns from data
Fixed responses	Dynamic responses
Easy to build	More complex
Best for FAQs	Best for conversations

✦ Part 15: Improving Rule-Based Chatbots

Instead of exact matching:

hello

Use keyword matching.

Example:

if "hello" in user:

Now it accepts:

hello

hello bot

hello sir

hello there

Part 16: Mini FAQ Chatbot

```
faq = {  
    "fees": "Course fee is ₹15,000.",  
    "duration": "Course duration is 3 months.",  
    "location": "We are located in Kochi.",  
    "contact": "Call us at 9876543210."  
}
```

```
while True:
```

```
    user = input("You: ").lower()
```

```
    if user == "bye":  
        print("Bot: Thank you!")  
        break
```

```
    found = False
```

```
    for key in faq:
```

```
        if key in user:  
            print("Bot:", faq[key])
```

```
            found = True
```

```
            break
```

```
    if not found:
```

```
        print("Bot: Sorry! Please contact our office.")
```

Part 17: Real-World Applications

- College Admission Bot
 - Hospital Appointment Bot
 - Restaurant Menu Bot
 - Hotel Booking Assistant
 - Banking FAQ Bot
 - Customer Support Bot
 - HR Helpdesk Bot
-

Part 18: Theory Questions

What is a Rule-Based Chatbot?

A chatbot that responds using predefined rules and conditions.

Which Python statement is commonly used?

if-elif-else

Why is lower() used?

To ignore uppercase and lowercase differences.

Why use a dictionary?

To store questions and responses efficiently.

What happens if no rule matches?

The chatbot displays a default response.

Part 19: Interview Questions

Explain the working of a Rule-Based Chatbot.

It compares the user's input with predefined rules and returns the corresponding response.

Why is while True used?

To keep the chatbot running until the user exits.

Difference between exact matching and keyword matching?

Exact matching requires the complete input to match, while keyword matching checks whether a specific word exists in the sentence.

Can a Rule-Based Chatbot learn from users?

No. It only follows predefined rules.

Where are Rule-Based Chatbots used?

Customer support, FAQs, college enquiries, banking, hospitals, and websites.

Tasks for Day 59

Theory

1. Define Rule-Based Chatbot.
 2. Explain the working of a Rule-Based Chatbot.
 3. What is the purpose of lower()?
 4. Why is a dictionary useful in chatbots?
 5. List the advantages and limitations of Rule-Based Chatbots.
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Practical Task 1

Create a chatbot that responds to:

- Hello
- Hi
- Bye
- Your Name

Practical Task 2

Create a **College Enquiry Chatbot** with at least five questions.

Practical Task 3

Modify the chatbot to use a Python dictionary instead of multiple if-elif statements.

Practical Task 4

Add keyword matching so that the chatbot responds even if the user types a longer sentence.

Example:

Hello there

↓

Hello!

Practical Task 5

Add a default response for unknown questions.

Example:

Sorry, I don't understand your question.